

# TPS Commissioning

## Oral Exam Master Packet

Dose Calculation Algorithms, Beam Modeling, Validation, and Clinical Commissioning

### High-yield point

This document is written specifically for oral-board style questioning and TPS commissioning preparation. The emphasis is not simply memorizing definitions, but understanding:

- How dose calculation algorithms work
- How TPS beam models are created
- How heterogeneity corrections are handled
- Why Monte Carlo and LBTE-based algorithms differ from kernel-based approaches
- How TPS systems are commissioned and validated clinically
- How to troubleshoot beam-model discrepancies

### Common trap

The most common oral-board failure is describing algorithms superficially without understanding:

- radiation transport,
- heterogeneity handling,
- beam modeling physics,
- or why algorithms fail clinically.

## Contents

|           |                                        |           |
|-----------|----------------------------------------|-----------|
| <b>1</b>  | <b>What is TPS Commissioning?</b>      | <b>2</b>  |
| <b>2</b>  | <b>Dose Calculation Algorithms</b>     | <b>2</b>  |
| 2.1       | Categories of Algorithms . . . . .     | 3         |
| 2.2       | Pencil Beam Algorithms . . . . .       | 3         |
| 2.3       | Collapsed Cone Convolution . . . . .   | 3         |
| 2.4       | AAA . . . . .                          | 4         |
| 2.5       | Acuros XB . . . . .                    | 5         |
| 2.6       | Monte Carlo . . . . .                  | 5         |
| <b>3</b>  | <b>Dose-to-Water vs Dose-to-Medium</b> | <b>6</b>  |
| <b>4</b>  | <b>Beam Modeling</b>                   | <b>6</b>  |
| 4.1       | Beam Model Parameters . . . . .        | 6         |
| <b>5</b>  | <b>TPS Validation</b>                  | <b>7</b>  |
| 5.1       | Validation Measurements . . . . .      | 8         |
| <b>6</b>  | <b>CT-Density Tables</b>               | <b>8</b>  |
| <b>7</b>  | <b>IMRT vs VMAT Commissioning</b>      | <b>8</b>  |
| <b>8</b>  | <b>TPS QA</b>                          | <b>9</b>  |
| <b>9</b>  | <b>Optimization Algorithms</b>         | <b>9</b>  |
| <b>10</b> | <b>Final High-Yield Oral Questions</b> | <b>10</b> |
| <b>11</b> | <b>Final High-Yield Checklist</b>      | <b>10</b> |

# 1 What is TPS Commissioning?

## Definition

**TPS commissioning** is the process of configuring, modeling, validating, and clinically verifying a treatment planning system using measured machine data before patient treatments are performed.

## Oral-exam answer

### What is the goal of TPS commissioning?

The goal is to ensure that the TPS accurately models:

- beam output,
- depth dose,
- profiles,
- heterogeneity effects,
- MLC behavior,
- and dose delivery under clinical conditions.

Commissioning establishes confidence that calculated dose distributions represent physical reality within clinically acceptable tolerances.

## Clinical workflow

### Typical TPS commissioning workflow

1. Acceptance testing
2. Beam data acquisition
3. Data processing
4. Import into TPS
5. Beam model generation
6. Parameter tuning
7. Open field validation
8. Heterogeneity validation
9. IMRT/VMAT validation
10. End-to-end testing
11. Clinical release

# 2 Dose Calculation Algorithms

## 2.1 Categories of Algorithms

| Algorithm      | Method                        | Strengths                        | Weaknesses                 |
|----------------|-------------------------------|----------------------------------|----------------------------|
| Pencil Beam    | 1D kernel scaling             | Very fast                        | Poor in heterogeneity      |
| Collapsed Cone | Convolution/superposition     | Good balance                     | Approximate transport      |
| AAA            | Anisotropic convolution       | Fast and accurate                | Still approximate          |
| Acuros XB      | LBTE solver                   | Excellent heterogeneity accuracy | Slower                     |
| Monte Carlo    | Particle transport simulation | Gold standard                    | Statistical noise / slower |

Table 1: Comparison of common TPS dose calculation algorithms.

## 2.2 Pencil Beam Algorithms

### Definition

Pencil beam algorithms model radiation as a collection of narrow beamlets whose dose deposition kernels are summed together.

### Common trap

Pencil beam algorithms generally fail in:

- low-density lung,
- interfaces,
- small fields,
- regions with lateral electron disequilibrium.

### Oral-exam answer

#### Why do pencil beam algorithms fail in lung?

Pencil beam algorithms primarily correct dose using radiological depth scaling and do not accurately model lateral electron transport. In lung, electrons can travel long distances laterally, causing loss of charged particle equilibrium that PB algorithms cannot properly model.

## 2.3 Collapsed Cone Convolution

### Definition

Collapsed cone convolution-superposition (CCCS) is a kernel-based algorithm that transports energy deposition along discrete directions called cones.

### Key physics

**Basic idea of convolution/superposition**

$$D(\vec{r}) = \int T(\vec{r}') K(\vec{r} - \vec{r}') d\vec{r}'$$

where:

- $T$  = TERMA
- $K$  = dose deposition kernel

### Oral-exam answer

**What is TERMA?**

TERMA stands for:

Total Energy Released per unit MAss

It represents the initial energy transferred from photons to charged particles before electron transport occurs.

### Oral-exam answer

**How does collapsed cone handle heterogeneity?**

Collapsed cone algorithms scale and deform energy deposition kernels according to density. The kernels are transported along discrete ray directions, allowing approximate modeling of scatter transport in heterogeneous media.

## 2.4 AAA

### Definition

AAA (Anisotropic Analytical Algorithm) is a convolution-superposition algorithm that models dose using multiple photon and electron sources combined with anisotropic scatter kernels.

### High-yield point

AAA separates dose contributions into:

- primary photons,
- extra-focal photons,
- contaminant electrons.

### Oral-exam answer

**Why is AAA better than pencil beam?**

AAA models lateral electron transport and anisotropic scatter more realistically than PB algorithms, leading to improved accuracy in heterogeneous media and off-axis regions.

### Common trap

AAA can still struggle in:

- very small lung fields,

- interfaces,
- extreme heterogeneity,
- severe disequilibrium conditions.

## 2.5 Acuros XB

### Definition

Acuros XB is a deterministic radiation transport solver that numerically solves the Linear Boltzmann Transport Equation (LBTE).

### Key physics

#### Linear Boltzmann Transport Equation

The LBTE describes:

- particle interactions,
- scattering,
- attenuation,
- energy transport,
- angular transport.

### Oral-exam answer

#### How is Acuros different from Monte Carlo?

Monte Carlo statistically simulates individual particle histories using random sampling, whereas Acuros deterministically solves transport equations on a spatial grid.

### Oral-exam answer

#### Why is Acuros more accurate in lung?

Acuros explicitly models radiation transport physics and scatter in heterogeneous media rather than relying on density-scaled kernels.

## 2.6 Monte Carlo

### Definition

Monte Carlo algorithms simulate individual particle histories using probabilistic interaction physics.

### Oral-exam answer

#### What is a particle history in Monte Carlo?

A particle history represents the complete transport sequence of a particle:

- interactions,

- scattering,
- energy deposition,
- secondary particle production,
- until termination criteria are met.

### Common trap

Monte Carlo calculations contain statistical uncertainty.  
More particle histories reduce noise but increase computation time.

### Oral-exam answer

#### What are transport cutoffs?

Transport cutoffs define the minimum particle energy below which particle transport is terminated to reduce computation time.

## 3 Dose-to-Water vs Dose-to-Medium

### Key physics

$$D_w^m = D_m^m \times \frac{\left(\frac{S}{\rho}\right)_{\text{water}}}{\left(\frac{S}{\rho}\right)_{\text{medium}}}$$

### Oral-exam answer

#### What is the difference between dose-to-water and dose-to-medium?

Dose-to-medium represents energy deposited in the actual physical medium. Dose-to-water converts the result into an equivalent water dose using stopping power ratios.

### Common trap

Dose-to-water and dose-to-medium differences become more significant in:

- bone,
- high-Z materials,
- heterogeneous interfaces.

## 4 Beam Modeling

### 4.1 Beam Model Parameters

**High-yield point**

The TPS beam model is created by adjusting parameters representing:

- primary source size,
- energy spectrum,
- flattening filter,
- extra-focal radiation,
- electron contamination,
- MLC transmission,
- DLG,
- tongue-and-groove effects.

**Oral-exam answer**

**What affects penumbra width?**

Penumbra width is strongly affected by:

- effective source size,
- focal spot size,
- MLC modeling,
- jaw geometry.

**Oral-exam answer**

**What parameter would you investigate if shallow PDD does not match?**

Potential causes include:

- electron contamination modeling,
- beam energy spectrum,
- buildup modeling,
- measurement setup issues.

## 5 TPS Validation

**Common trap**

Validation must be independent of the data used to create the beam model.

**Oral-exam answer**

**How do you know the TPS is ready for validation?**

The TPS is generally considered ready when:



- profiles match measured data,
- PDDs are acceptable,
- output factors are acceptable,
- no major systematic trends remain,
- MLC behavior appears physically reasonable.

## 5.1 Validation Measurements

| Validation Test        | Purpose                         |
|------------------------|---------------------------------|
| Point dose             | Absolute dose accuracy          |
| PDD comparison         | Depth dose accuracy             |
| Profiles               | Penumbra / flatness / symmetry  |
| Heterogeneity testing  | Transport accuracy              |
| Small field validation | MLC and disequilibrium accuracy |
| IMRT QA                | Modulation accuracy             |
| VMAT QA                | Dynamic delivery validation     |

Table 2: Typical TPS validation measurements.

## 6 CT-Density Tables

### Oral-exam answer

#### How do you generate a CT-density table?

A CT-density calibration phantom containing known density inserts is scanned. Hounsfield Units are measured and mapped to electron density or mass density for TPS dose calculations.

### Common trap

Different CT protocols or kVp settings may require different CT-density tables.

## 7 IMRT vs VMAT Commissioning

### Oral-exam answer

#### What requires extra attention in VMAT commissioning?

VMAT commissioning requires additional validation of:

- gantry speed,
- dose rate modulation,
- MLC motion,

- control point interpolation,
- rotational delivery accuracy.

## 8 TPS QA

### Oral-exam answer

#### How often should TPS QA be performed?

TPS QA should be performed:

- after upgrades,
- after beam model changes,
- periodically according to institutional policy,
- after major hardware/software modifications.

### High-yield point

Common TPS QA references:

- TG-53
- MPPG 5.a
- TRS-430
- TG-244

## 9 Optimization Algorithms

### Definition

Inverse planning optimization attempts to minimize an objective function while satisfying dose constraints.

### Oral-exam answer

#### Common optimization methods

- Gradient descent
- Simulated annealing
- Aperture optimization
- Stochastic optimization

## 10 Final High-Yield Oral Questions

### Oral-exam answer

**What is the biggest weakness of kernel-based algorithms?**

Kernel-based algorithms approximate electron transport and may fail in severe heterogeneity or disequilibrium conditions.

### Oral-exam answer

**What is the gold standard dose calculation algorithm?**

Monte Carlo is generally considered the gold standard because it directly simulates particle transport physics.

### Oral-exam answer

**Why is validation independent from commissioning?**

Validation ensures the TPS accurately predicts scenarios that were not used to generate the beam model.

## 11 Final High-Yield Checklist

### High-yield point

You should be able to explain:

1. How PB, CCC, AAA, Acuros, and MC work
2. How heterogeneity is handled
3. Dose-to-water vs dose-to-medium
4. How beam models are created
5. What DLG represents
6. How to troubleshoot beam-model problems
7. How TPS validation is performed
8. TG-53 vs MPPG 5.a philosophy
9. IMRT vs VMAT commissioning
10. Monte Carlo limitations

### Common trap

The single best oral-board sentence:

*A TPS is only as accurate as its beam model, commissioning measurements, and validation methodology.*